

Lesson 2: Integrating the equations of motion

This section will focus on finding the motion of a particle when the acceleration is a given function of time $a = a(t)$, or position $a = a(x)$, or velocity $a = a(v)$. We will look at these three cases and how to solve them.

1 Case I: The acceleration is a function of time

To find the velocity and position when the acceleration is provided as $a = a(t)$, we integrate the relation

$$\begin{aligned}\frac{dv}{dt} &= a \implies dv = a(t)dt \\ \int_{v_0}^v dv &= \int_{t_0}^t a(t)dt \implies v - v_0 = \int_{t_0}^t a(t)dt\end{aligned}$$

$$v(t) = v_0 + \int_{t_0}^t a(t)dt$$

(1)

After we determine the function for velocity we can integrate $dx = vdt$ to find the position.

$$x(t) = x_0 + \int_{t_0}^t v(t)dt \quad (2)$$

Sample Problem 2.1

A motorcycle accelerates with a non-constant function of time: $a(t) = 2t \text{ m/s}^2$ between a time of $t = 2$ and $t = 4$ seconds. If we know that the motorcycle's velocity is $v = 20 \text{ m/s}$ at $t = 2$ how fast is it going at $t = 4$ seconds? How far did it travel during this two second interval?

We first determine the equation for velocity using eq. 1.

$$\begin{aligned} v(t) &= v_0 + \int_{t_0}^t a(t)dt = 20 + \int_2^t 2t dt \\ v(t) &= 20 + t^2 \Big|_2^t = 20 + t^2 - 2^2 \\ v(t) &= 16 + t^2 \end{aligned}$$

We can now solve for the speed at $t = 4$

$$v(t = 4) = 16 + (4)^2 = 32 \text{ m/s}$$

To find the position as a function of time, use eq. 2. I've chosen to set the starting position to be zero (*i.e.* $x_0 = 0$) at $t = 2$.

2.1 – cont

$$x(t) = x_0 + \int_{t_0}^t v(t) dt$$

$$x(t) = 0 + \int_2^t (16 + t^2) dt$$

$$x(t) = \left(16t + \frac{1}{3}t^3 \right) \Big|_2^t$$

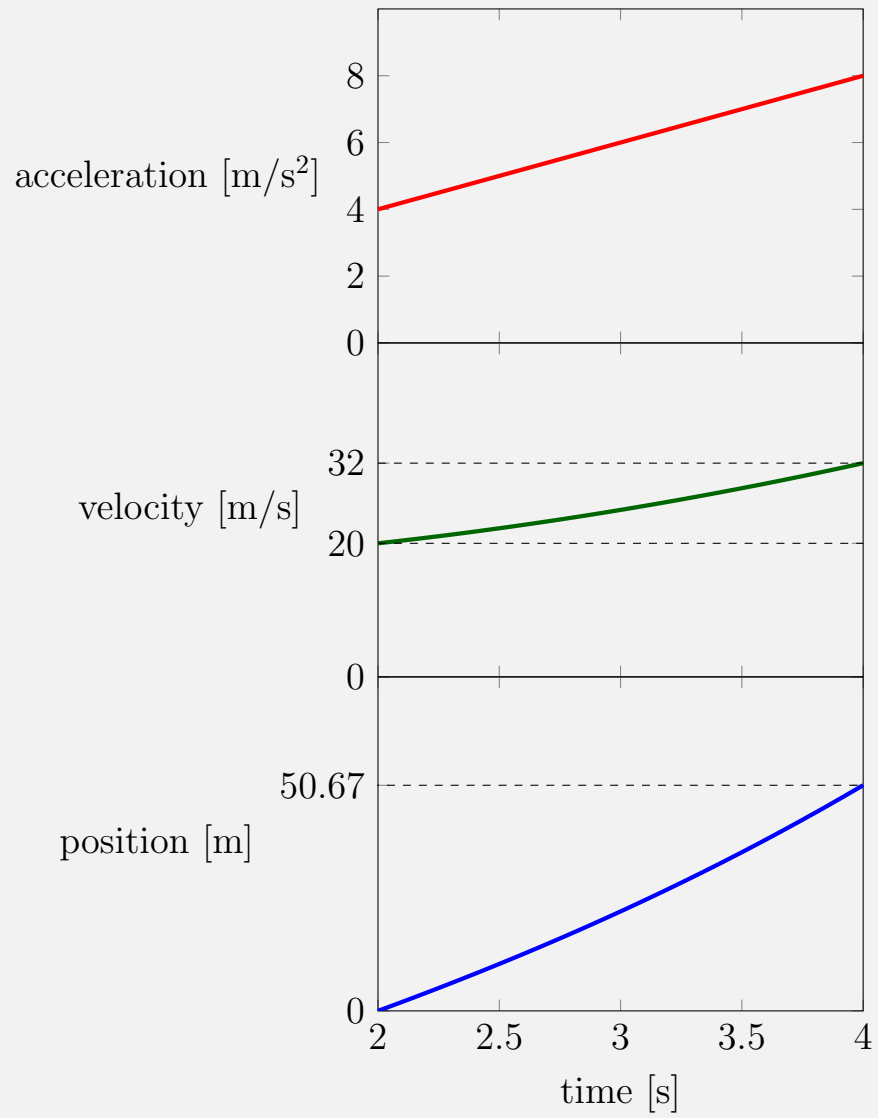
$$x(t) = \left(16t + \frac{1}{3}t^3 \right) - \left(16(2) + \frac{1}{3}2^3 \right)$$

$$x(t) = \frac{1}{3}t^3 + 16t - 34.67$$

The distance traveled at $t = 4$ is

$$x(t = 4) = \frac{1}{3}(4)^3 + 16(4) - 34.67 = 50.67 \text{ m}$$

2.1 – cont



2 Case II: The acceleration is a function of position

If the acceleration is specified as a function of position $a = a(x)$ we start with the relation

$$a = v \frac{dv}{dx}$$

This can be rearranged and integrated

$$\begin{aligned} v dv &= a dx \\ \int_{v_0}^v v dv &= \int_{x_0}^x a(x) dx \\ \left. \frac{v^2}{2} \right|_{v_0}^v &= \int_{x_0}^x a(x) dx \\ \frac{v^2}{2} - \frac{v_0^2}{2} &= \int_{x_0}^x a(x) dx \end{aligned}$$

resulting in the following expression that allows us to find v in terms of x .

$$\boxed{v^2 = v_0^2 + 2 \int_{x_0}^x a(x) dx} \quad (3)$$

To solve for the position, we use the relation $dt = \frac{dx}{v}$ and integrate

$$\boxed{t = t_0 + \int_{x_0}^x \frac{dx}{v(x)}} \quad (4)$$

Sample Problem 2.2

A car traveling at 60 mph collides with a crash barrier on the highway. After hitting the barrier, the car decelerates at a rate proportional to the distance x the car has moved into the barrier. The acceleration is given as $a = -16x$, where a and x are in ft/s^2 and ft , respectively. How far does the car travel into the barrier before coming to rest? How long does it take to come to rest?



In this problem since the acceleration is given as a function of position we start with eq. 3. I've chosen $x_0 = 0, t = 0$ to be the position/time of initial impact. Also, $60 \text{ mph} = 88 \text{ fps}$.

2.2 – cont

$$\begin{aligned}v^2 &= v_0^2 + 2 \int_{x_0}^x a(x) dx \\v^2 &= (88)^2 + 2 \int_0^x (-16x) dx \\v^2 &= 7744 - 32 \int_0^x x dx = 7744 - 32 \left. \frac{x^2}{2} \right|_0^x \\v^2 &= 7744 - 16x^2 \implies v(x) = \sqrt{7744 - 16x^2}\end{aligned}$$

The car comes to rest when $v = 0$:

$$0 = \sqrt{7744 - 16x^2} \implies x = 22$$

Technically $x = -22$ is also a solution to the above equation, but this is nonphysical as the barrier is located at $x \geq 0$.

We next want to find the position as a function of time by using eq. [4](#)

$$\begin{aligned}t &= t_0 + \int_{x_0}^x \frac{dx}{v(x)} \\t &= 0 + \int_0^x \frac{dx}{\sqrt{7744 - 16x^2}} \\t &= \frac{1}{4} \arcsin \left(\frac{x}{22} \right)\end{aligned}$$

At $x = 22$ ft the time is $t = \pi/8 \approx 0.393$ s.

3 Case III: The acceleration is a function of velocity

Our final case is when a is specified as a function of velocity, $a = a(v)$. We start with the relation

$$a = \frac{dv}{dt}$$

This can be rearranged and integrated

$$\begin{aligned} dt &= \frac{dv}{a(v)} \\ \int_{t_0}^t dt &= \int_{v_0}^v \frac{dv}{a(v)} \\ t - t_0 &= \int_{v_0}^v \frac{dv}{a(v)} \end{aligned}$$

resulting in the following expression that allows us to solve for v as a function of t .

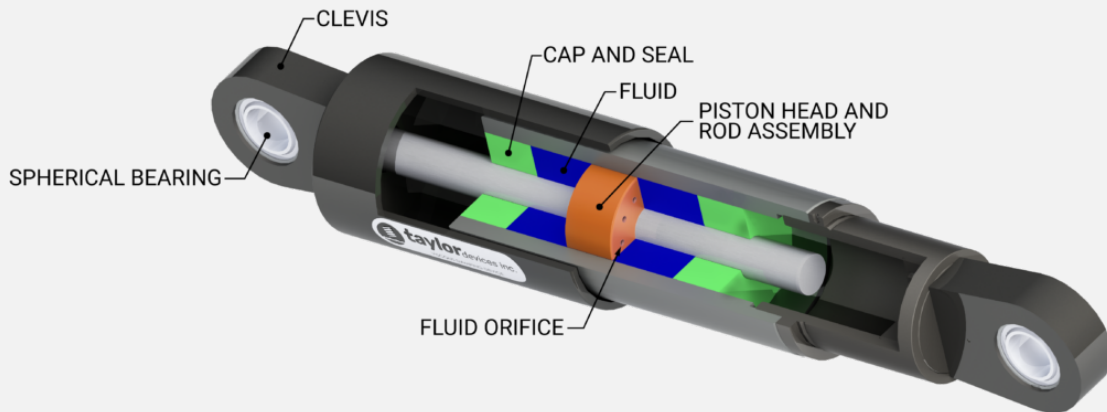
$$\boxed{t = t_0 + \int_{v_0}^v \frac{dv}{a(v)}} \quad (5)$$

Once we determine the functional form of $v(t)$ we can integrate it to determine $x(t)$. If instead we wish to find v in terms of x we start with $a = v \frac{dv}{dx}$ rearrange and integrate to find

$$\boxed{x = x_0 + \int_{v_0}^v \frac{v dv}{a(v)}} \quad (6)$$

Sample Problem 2.3

A linear damper utilizes a piston with orifices that moves through a fluid-filled cylinder. If the piston is given some initial velocity, it decelerates at a rate proportional to the velocity, $a = -kv$.



A testing rig starts the piston at $t = 0$ with $v = 30$ in/s. The piston is observed to have a velocity $v = 10$ in/s at time $t = 2.0$ s. Determine (a) the velocity of the piston at $t = 1.0$ s, (b) the time required for the shaft to come to rest, (c) the maximum travel of the piston, (d) the velocity when the piston reaches half its maximum travel.

The acceleration is specified as a function of velocity. We therefore start with eq. 5

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$$\begin{aligned}t &= t_0 + \int_{v_0}^v \frac{dv}{a(v)} \\t &= 0 + \int_{30}^v \frac{dv}{-kv} \\t &= -\frac{1}{k} \ln(v) \Big|_{30}^v = -\frac{1}{k} [\ln(v) - \ln(30)] \\-kt &= \ln\left(\frac{v}{30}\right) \\v &= 30e^{-kt}\end{aligned}$$

We can now solve for k by using the information that at $v(t = 2) = 10$.

$$\begin{aligned}-kt &= \ln\left(\frac{v}{30}\right) \\-k(2) &= \ln\left(\frac{10}{30}\right) = -1.1 \longrightarrow k = 0.549 \text{ s}^{-1}\end{aligned}$$

To find the velocity at any other time we use

$$\begin{aligned}v(t) &= 30e^{-0.549t} \\v(t = 1) &= 30e^{-0.549} = 17.3 \text{ in/s}\end{aligned}$$

Since the velocity decays exponentially the piston only comes to rest in the limit that $t \rightarrow \infty$. To find the maximum travel we find the

2.3 – cont

position as a function of time and take the long-time limit.

$$\begin{aligned}v &= \frac{dx}{dt} \implies dx = v dt \\dx &= 30e^{-0.549t} dt \\ \int_0^x dx &= \int_0^t 30e^{-0.549t} dt = \frac{30 \text{ in/s}}{-0.549 \text{ s}^{-1}} e^{-0.549t} \Big|_0^t \\x &= -54.6 \text{ in} (e^{-0.549t} - 1) \\x(t \rightarrow \infty) &= 54.6 \text{ in}\end{aligned}$$

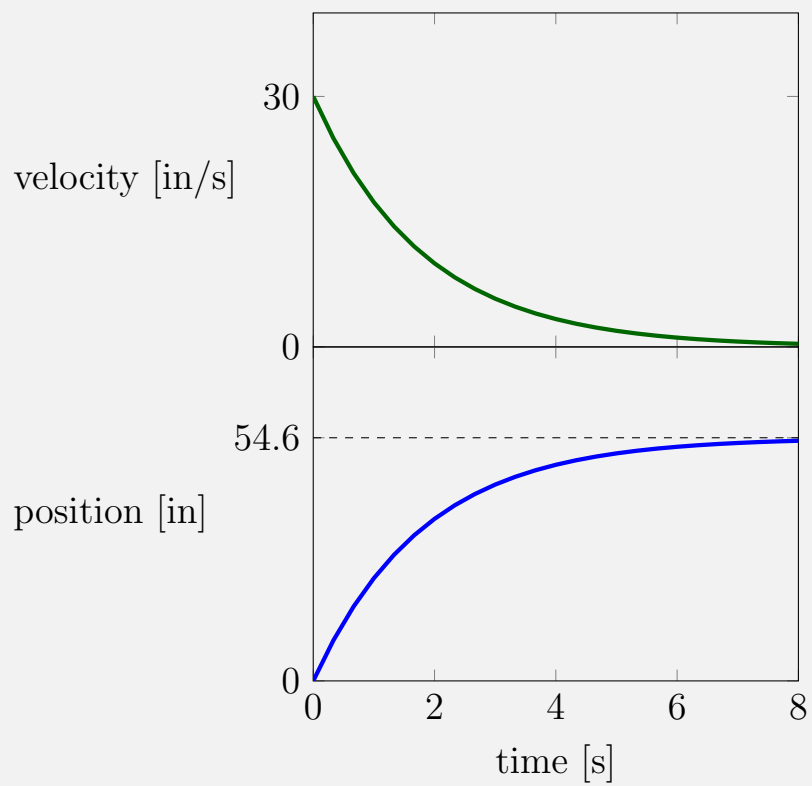
For part (d) we are asked to find the velocity when the piston is at half its travel, or at $x = 27.3$ in. There are two ways to do this. The first is to find the time when $x = 27.3$ in and then compute $v(t)$. I'll take another approach and use eq. 6.

$$\begin{aligned}x &= x_0 + \int_{v_0}^v \frac{v dv}{a(v)} \\x &= 0 + \int_{v_0}^v \frac{v dv}{-kv} \\x &= -\frac{1}{k} \int_{v_0}^v dv = -\frac{1}{k} v \Big|_{v_0}^v = -\frac{1}{k} (v - v_0)\end{aligned}$$

Solving the above for $v(x)$ we find

$$\begin{aligned}v(x) &= v_0 - kx \\v(x) &= 30 - 0.549x \\v(x = 27.3) &= 30 - 0.549(27.3) = 15 \text{ in/s}\end{aligned}$$

2.3 – cont

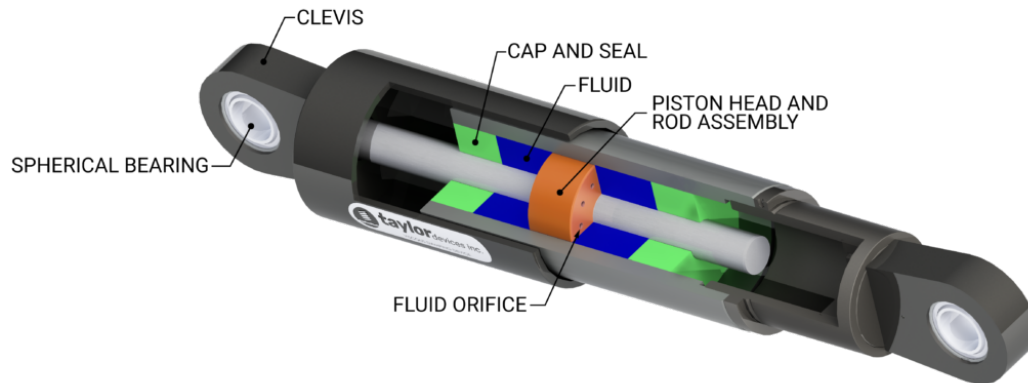


Sample Problems

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2. A car traveling at 60 mph collides with a crash barrier on the highway. After hitting the barrier, the car decelerates at a rate proportional to the distance x the car has moved into the barrier. The acceleration is given as $a = -16x$, where a and x are in ft/s^2 and ft , respectively. How far does the car travel into the barrier before coming to rest? How long does it take to come to rest?



3. A linear damper utilizes a piston with orifices that moves through a fluid-filled cylinder. If the piston is given some initial velocity, it decelerates at a rate proportional to the velocity, $a = -kv$.



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